

A Conjecture on the Existence of Primes in Intervals Between Consecutive Squares

Mohamed Amine Belachhab

Abstract

We present a conjecture concerning the existence of prime numbers in specific arithmetic progressions contained within intervals of the form $((n + 1)^2, (n + 2)^2)$. The conjecture provides an explicit construction that, if proven, would imply Legendre's conjecture on the existence of primes between consecutive perfect squares. We formulate the problem in terms of parametric families indexed by integers k , distinguishing between the cases where n is odd or even.

1. Introduction

The distribution of prime numbers between consecutive perfect squares has been a subject of considerable interest in number theory. Legendre's conjecture asserts that for every positive integer n , there exists at least one prime number in the interval $(n^2, (n + 1)^2)$. Despite substantial computational evidence supporting this claim, a proof remains elusive.

In this paper, we propose a constructive approach to this problem by identifying specific arithmetic progressions within the intervals $((n + 1)^2, (n + 2)^2)$ that we conjecture must contain at least one prime number. Our formulation provides an explicit parametric family indexed by a parameter k whose range depends on the parity of n .

2. Statement of the Conjecture

Conjecture 1. Let $n \geq 1$ be a positive integer. Then at least one of the following holds:

- (i) If n is odd, there exists $k \in \{1, 2, \dots, n\}$ such that $n^2 + 2(k + n + 1)$ is prime.
 - (ii) If n is even, there exists $k \in \{0, 1, \dots, n - 1\}$ such that $n^2 + 2(k + n + 1) + 1$ is prime.
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3. Properties and Observations

3.1. Range of Generated Values

We first verify that the values generated by the conjecture lie within the appropriate interval.

Proposition 2. The integers generated by Conjecture 1 lie in the open interval $((n + 1)^2, (n + 2)^2)$.

Proof. For odd n with $k \in \{1, \dots, n\}$:

- When $k = 1$: $n^2 + 2(1 + n + 1) = n^2 + 2n + 4 > (n + 1)^2 = n^2 + 2n + 1$.

- When $k = n$: $n^2 + 2(n + n + 1) = n^2 + 4n + 2 < (n + 2)^2 = n^2 + 4n + 4$.

For even n with $k \in \{0, \dots, n - 1\}$:

- When $k = 0$: $n^2 + 2(n + 1) + 1 = n^2 + 2n + 3 > (n + 1)^2$.
- When $k = n - 1$: $n^2 + 2(2n) + 1 = n^2 + 4n + 1 < (n + 2)^2$.

Thus all generated values lie in $((n + 1)^2, (n + 2)^2)$. $\square$

3.2. Arithmetic Progression Structure

The conjecture asserts the existence of primes in specific arithmetic progressions.

Observation 3. The values in Conjecture 1 form arithmetic progressions of odd integers with common difference 2.

For odd n , the sequence $\{n^2 + 2(k + n + 1) : k = 1, \dots, n\}$ consists of n consecutive odd integers.

For even n , the sequence $\{n^2 + 2(k + n + 1) + 1 : k = 0, \dots, n - 1\}$ consists of n consecutive odd integers.

In both cases, we are examining approximately n consecutive odd numbers in an interval of length approximately $2n$ centered around $(n + \frac{3}{2})^2$.

3.3. Connection to Legendre's Conjecture

Proposition 4. Conjecture 1 implies Legendre's conjecture.

Proof. Legendre's conjecture asserts that for every $n \geq 1$, there exists a prime in $(n^2, (n + 1)^2)$. By Proposition 2, Conjecture 1 guarantees a prime in $((n + 1)^2, (n + 2)^2)$. Replacing n with $n - 1$, we obtain a prime in $(n^2, (n + 1)^2)$, thus establishing Legendre's conjecture. $\square$

4. Heuristic Considerations

While we do not provide a proof, we offer heuristic reasoning supporting the conjecture.

By the Prime Number Theorem, the density of primes near a number m is approximately $1 / \ln m$. In our case, near $(n + 1)^2$, this density is approximately $1 / (2 \ln n)$.

Among n odd integers, the expected number of primes is heuristically:

$$\frac{n}{2 \ln n}.$$

For $n \geq 4$, this quantity exceeds 1, suggesting that we should expect at least one prime in the progression. However, this heuristic argument does not constitute a proof, as the distribution of primes in short intervals is not fully understood without assuming results such as the Riemann Hypothesis.

5. Relation to Known Results

The conjecture is closely related to several open problems in analytic number theory:

1. **Legendre's Conjecture:** As established in Proposition 4, our conjecture implies Legendre's conjecture.
2. **Prime gaps:** The conjecture concerns the existence of primes in intervals of length $O(n)$ starting near n^2 , which is related to bounds on prime gaps.
3. **Primes in arithmetic progressions:** Our conjecture can be viewed as asserting that certain short arithmetic progressions of odd numbers must contain primes.

Proving the conjecture would require techniques capable of establishing the existence of primes in short intervals or arithmetic progressions, potentially involving sieve methods or explicit estimates from analytic number theory.

6. Conclusion

We have formulated a precise conjecture regarding the existence of primes in specific arithmetic progressions within intervals between consecutive perfect squares. The conjecture, if proven, would resolve Legendre's conjecture and contribute to our understanding of the fine-scale distribution of prime numbers.

The explicit constructive nature of our formulation may provide a pathway for future investigations using elementary, sieve-theoretic, or analytic methods.

References

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